

Figure 1: A typical process control network

Honeypots and honeynets

Wikipedia defines a honeypot as ‘a trap set to detect, deflect, or in some manner counteract attempts at unauthorised use of information systems.’ Generally, it consists of a computer, data or a network site that appears to be part of a network, but is actually isolated and monitored, and which seems to contain information or a resource of value to attackers. Thus, an attacker may attack a SCADA honeypot perceiving it to be a true SCADA system. Multiple honeypots configured to mimic various devices or operating systems is a honeynet. Depending upon the requirement, honeypots and honeynets can be deployed at any of the following locations:

- Directly accessible from the Web
 - In a de-militarised zone where access is allowed from the Internet as well as from the protected internal network
 - On the internal network
- Honeypots and honeynets help to ensure security in various ways:
- They divert the attacker’s attention to an easy target rather than the actual system.
 - Log the attackers’ activities for further analysis to gain in-depth knowledge about the attack and to develop prevention techniques.
 - Provide forensic information, which is required by law enforcement agencies to establish that an attack occurred.

Characteristics of honeypots and honeynets

- They look ‘genuine’, exactly like the system they mimic—an attacker should not be able to make out that they are modified systems.
- Allow controlled traffic towards the Internet—an attacker should not be able to use the honeypot as a stepping stone for further attacks on the Internet.
- May contain dummy information, for example a SCADA honeypot may contain a Web page resembling the genuine SCADA system. This will attract the attackers and keep them engaged, ultimately resulting in more time and attack techniques being used on this system.

Honeyd: An open source honeypot

As defined by honeyd.org, Honeyd is a small daemon

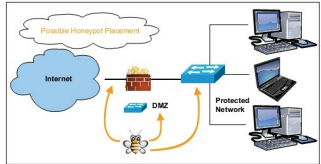


Figure 2: Honeypot placement

that creates virtual hosts on a network. These hosts can be configured to run arbitrary services, and their personality can be adapted so that they appear to be running certain operating systems. Honeyd enables a single host to claim multiple addresses (tested up to 65536) on a LAN for network simulation. Honeyd improves cyber security by providing mechanisms for threat detection and assessment. It also deters adversaries by hiding real systems in the middle of virtual systems.

The Honeyd configuration file defines how the configured honeypot will respond to various types of requests such as ICMP Ping, requests on UDP ports, TCP SYN, etc, thus, in a way, defining the status of various ports and services. This reply is interpreted by the scanning tool as a system running a corresponding service.

The basics of nmap port scanning

Let us understand the process of port scanning that is used by the network scanning tool, nmap. A typical SYN scan of nmap sends a SYN packet to the destination IP address on the port number to be scanned. There are three possible replies from the destination IP as shown in Table 1.

Table 1: nmap port status

No	Response	Port status	Explanation
1	SYN ACK	Open	Service running on the port
2	RST	Closed	No service running on the port
3	No response	Filtered	Could not determine port status

Installation

The simplest way to install honeyd under Ubuntu 12.0.4 is to use the following command:

```
sudo apt-get install honeyd
```

Honeyd is installed in `/usr/share/honeyd`. Once installed, it can be configured to mimic various operating systems which appear to run with various services. First, let us understand